AI-Driven Human–Exoskeleton Digital Twin: progress report 2

11 November 2026 · Kurian Jose and Jasith Kadian · work on the mentor's ExoGait plan of 14 October

This follows the report of 10 November, which covered the simulation half. The AI half is now done too, and the two are joined.

Summary

- **The preprocessing pipeline is built and run on all 19 participants (RO1).** 5,964 clean strides, each on 101 points, with a log of every stride that was dropped and why.
- **The torque models are trained and scored on held-out data (RO2).** A personalised LSTM predicts the torque the exoskeleton delivers at the joint to 2.4 Nm in walking (R^2 0.97 at the hip, 0.94 at the knee). For a person it has never seen, the error is 4.0 Nm.
- **H1, H2 and H3 are supported.** Personalised models beat population models in all 19 participants. Gait phase and locomotion mode both help, but only a little once the model is strong.
- **A finding outside the plan: joint acceleration is the input that matters for the motor torque.** It follows from last week's result that half the motor torque accelerates the motor's own rotor. Adding it nearly halves a linear model's error.
- **The AI torque works in the simulation (RO4).** Replayed in the musculoskeletal model in place of the measured torque, the LSTM's torque gives the same muscle effort (−1.4%, not significant).
- **A new check of the simulation against a measurement it never saw.** The exoskeleton's footplates record how the load is split between the feet. The model reproduces that split (r = 0.90), and the comparison shows where its foot contact is wrong.
- **The model is now scaled to each participant.** Every participant's height and weight are published in the dataset's paper, which we had missed. On bodies scaled to them, the assistance result is unchanged: the optimised torque still lowers effort by 34% (16 of 16).
- **EMG (RO5, H5) is still blocked.** The paper says EMG was recorded for five participants; it was not released. We wrote to the authors on 11 November.

Status against the plan

Plan item	Status	Where it stands
Layer 1: ExoGait data	Done	19 participants checked, column by column
Layer 2: processing (RO1)	**Done**	Strides, stride features, exclusion log, whole-block train/validation/test splits, a leakage check
Layer 3A: AI torque prediction (RO2, H1–H3)	**Done**	Five models, two schemes, four input sets, two torques; scored once on the test blocks
Layer 3B: OpenSim model (RO3)	Done	X2 model; angle mapping checked on all 19
Layer 4: personalised twin state	**Done for walking**	A generic body, and one scaled to each participant's height and mass
Layer 5: Moco optimal control	Done for walking	38 of 38 tracking solves converge; load split checked against the footplates
Layer 6: assistance, H4 (RO4)	**Done on mean strides**	Exo off, measured, AI-predicted and optimised torque compared
EMG-informed modulation (RO5, H5)	Blocked	Authors asked for the EMG of five participants
Evaluation at scale (RO6)	Partly	Prediction and mean strides done; single strides and ramps not started

Results

1. Preprocessing (RO1)

Each recording is cut into strides at the heel strikes the dataset marks, and each stride is resampled to 101 points. Of 9,697 strides detected, 5,964 are kept. The rest cross into a turn or a transition (2,237), are marked incomplete by the dataset itself (1,399), have an outlier duration (75) or have gaps in sampling (22). Data is split by whole passes, never by random strides, because neighbouring strides are near copies of each other.

Two faults in the data were found on the way. One participant's left-knee torque sensor is stuck at a constant value for a third of their session. In five files the clock restarts and runs over the same seconds twice.

2. Torque prediction (RO2)

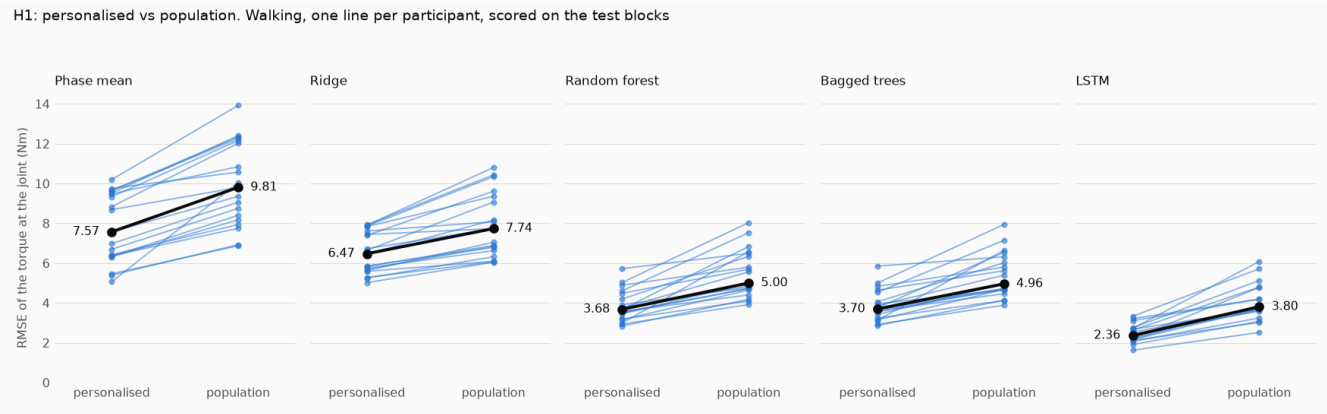
Five models were compared: the average torque at each point of the stride ("phase mean"), ridge regression, a random forest, bagged trees and an LSTM. Each was trained two ways: on the participant's own earlier passes (personalised), and on the other 18 participants only (population). All were scored on passes held out until the end.

Error in walking, in Nm, for the torque delivered at the joint:

Model	Personalised	Population (unseen person)
Phase mean	7.7	9.9
Ridge	6.5	7.9
Random forest	3.8	5.5
LSTM	**2.4**	**4.0**

The torque itself ranges over roughly ±40 Nm, so the personalised LSTM's error is about 3% of the range. On ramps it is 3.0 Nm and on stairs 3.0 to 4.0 Nm. The motor's torque is harder to predict (4.8 Nm personalised).

3. The three prediction hypotheses



Each line is one participant. Personalised is lower for every participant in every model.

Hypothesis	Test	Result
H1: personalised beats population	LSTM, walking	2.36 vs 3.80 Nm; 19 of 19 participants; p = 0.000004
H2: gait phase helps	Random forest, walking	3.68 vs 4.02 Nm without it; 19 of 19; p = 0.000004
H3: locomotion mode helps	LSTM, all five modes	2.81 vs 2.95 Nm without it; 18 of 19; p = 0.000008

All three hold. H2 and H3 are small for the strong models: a forest or an LSTM that sees both legs' angles, their velocities and the foot-load signal can work out the phase and the mode for itself. For the simple phase-mean model, phase is worth 4 Nm.

4. Joint acceleration

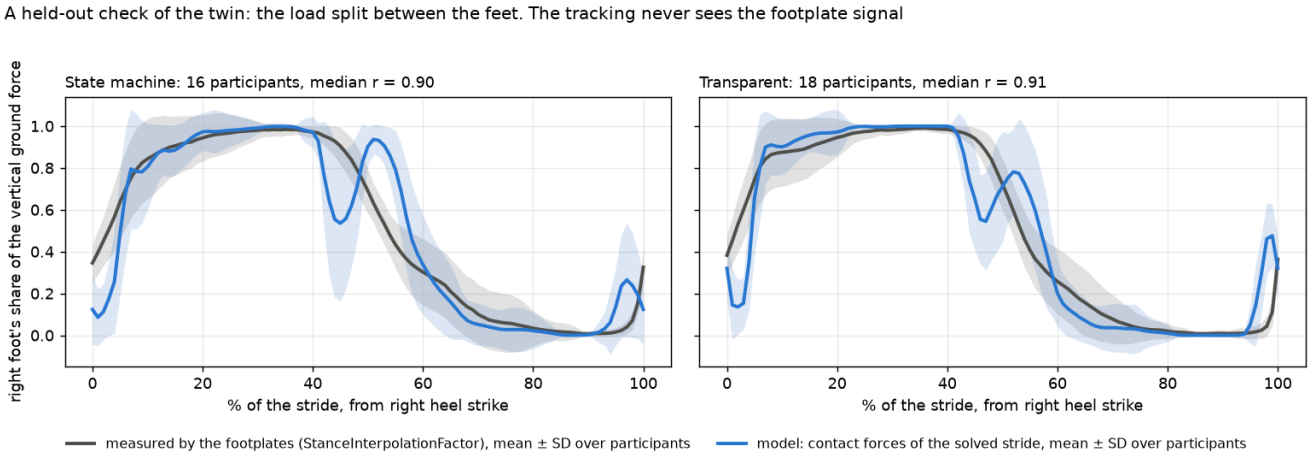
The first report showed that the motor's torque is the joint's torque plus what it takes to accelerate the motor's own rotor. So a model that sees only angle and velocity at one instant cannot predict the motor torque. Adding the acceleration (computed from the recorded velocity, so nothing new is measured) takes ridge's walking error on the motor torque from 16.0 to 9.2 Nm and the random forest's from 9.9 to 6.1 Nm, in all 19 participants. The LSTM gains least (5.3 to 4.9 Nm), because it already sees how the velocity changes along the stride.

5. The AI torque in the simulation (RO4)

The LSTM's predicted torque for each participant's held-out strides was averaged and replayed in the musculoskeletal model in place of the measured torque. The predicted average is 1.2 Nm off the measured one. The model's muscle effort with it differs from the effort with the measured torque by −1.4% (median of 15 participants, range −4.2% to +5.4%, not significant). One participant's solve missed our accuracy check and is left out.

So the four assistance cases now stand side by side: exoskeleton off, measured, AI-predicted, and optimised. The optimised torque still lowers effort by 34% against the measured one at the same torque budget (16 of 16).

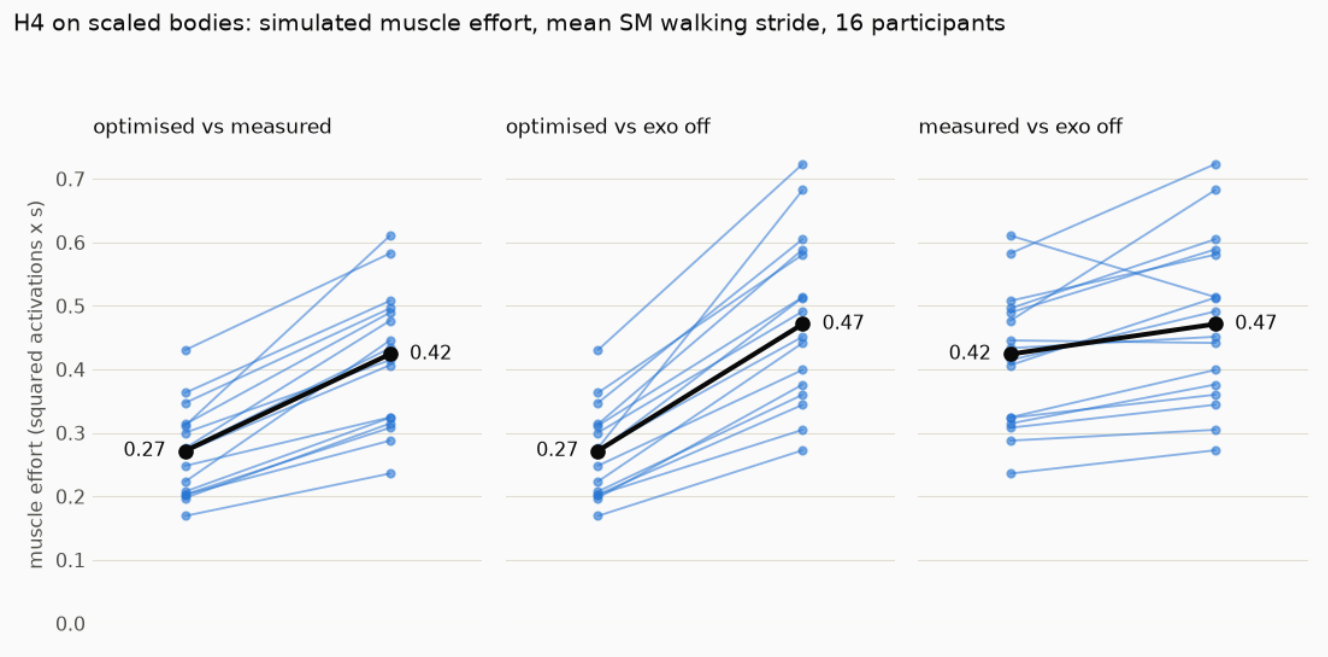
6. A check of the simulation against the footplates



The dataset has no ground reaction forces, which is the usual way to check a walking simulation. But one of its columns turns out to be the share of the body's weight on one foot, measured by force sensors in the exoskeleton's footplates. The simulation is never shown this signal, so it is a fair test of the forces the model invents under its feet.

The model follows it (r = 0.90, median of 34 solves). It gets both single-support phases right. It gets the hand-over between the feet wrong: the real load moves across smoothly, while the model's incoming foot takes the load, gives it back and takes it again. That is a known weakness of simple foot-contact models, now measured for this one.

7. The model scaled to each participant



Until now one 62 kg body stood in for 19 people of 47 to 90 kg. We scaled the model to each participant's published height and weight (the exoskeleton stays at 20.7 kg) and solved the walking and the assistance cases again.

	Generic body	**Scaled body**
Walking simulations passing our checks	34 of 38	35 of 37
Effort, optimised against measured torque	0.268 vs 0.438; 16 of 16	0.272 vs 0.424; 16 of 16
Median saving from the optimised torque	33.6%	33.9%
Measured torque against exoskeleton off	lower in 14 of 16; p = 0.011	lower in 14 of 16; p = 0.002
Load split against the footplates (r)	0.90	0.91

So H4 does not depend on the generic body. Scaling also moves each person's effort the way their weight says it should: the lightest participant (51 kg) works 29% harder than on the generic body and the heaviest (90 kg) 20% less (rank correlation with body mass −0.63, p = 0.011). The exoskeleton weighs the same whoever wears it, so it is a larger share of a light person's load.

The 16 participants are not quite the same 16: one enters and one leaves when the body is scaled. The AI-torque case has not been run on the scaled bodies yet.

Limits

- **Simple scaling.** One length factor from height, because no segment lengths were measured. Muscle effort is a model quantity, not checked against oxygen use or EMG.
- **Offline models.** Every torque model works on whole strides, which needs the next heel strike. They are not yet usable in a live controller.
- **Same-session test data.** The held-out passes come from the same session as the training passes, not another day.
- **Mean strides, walking only,** in the simulation. Ramps and stairs are in the prediction results but not in the simulation.
- **The solver's answer depends on its starting point.** The measured, AI and exo-off cases were each solved from two starts; a wider search is still to do.

What we need

1. **A decision on EMG.** If the authors do not release it by 1 December, we propose to test H5 on the model's own muscle activity instead, stated as such, and to report H5 on real EMG as not testable.
2. **DGX time** for the full evaluation on single strides: about 5,000 simulations.
3. **Your view on the two torques.** We predict both and replay the torque at the joint. If only one is wanted, we propose that one.

Next

1. Single strides and ramps in the simulation.
2. The AI-torque case on the scaled bodies.
3. The write-up of methods and results.